

Charotar University of Science and Technology

CHARUSAT Campus, Changa – 388 421.

M Sc (Nano Science and Technology)
SEMESTER II EXAMINATION 2011
NIT 711 - Elective I: Biophysical Principles

Date: 6th May 2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.
4. Use of cell phones is strictly prohibited.
5. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 20

Q1. Choose the correct option and put $\sqrt{\quad}$ mark in front of it:

1. Diffusion rate is inversely proportional to viscosity. True / False
2. The ratio of the diffusion constant to the unknown thickness of the membrane is known
(a) Permeability Ratio (b) Partition coefficient (c) Permeability coefficient
(d) Permeability constant
3. Greater the molecular weight of the solute, smaller is the permeability coefficient.
True / False
4. Graham's law of diffusion relates
(a) Rate of diffusion and density (b) Rate of diffusion and viscosity
(c) Diffusion constant and diffusion (d) Rate of diffusion and permeability
5. The plasma membranes of living cells are
(a) Selectively semipermeable (b) Permeable (c) Selectively permeable
(d) Semipermeable
6. Which of the following best describes osmosis?
(a) Spontaneous passage of solvent across a semipermeable membrane
(b) Spontaneous passage of solvent across a permeable membrane
(c) Spontaneous passage of solute across a semipermeable membrane
(d) Spontaneous passage of solute across a permeable membrane
7. Which of the following is NOT true for active transport?
(a) It does not require carrier protein (b) It requires energy input
(c) It is responsible for Na and K ion transport across the membrane
(d) None of the above
8. Abbe-Nollet's experiment is used to demonstrate
(a) Diffusion (b) Reverse Osmosis (c) Membrane permeability (d) Osmosis
9. Hydrostatic pressure in glomerular capillaries remains constant while it declines in other capillaries. True/False

-
10. Which of the following is NOT TRUE for capillary filtration?
(a) occurs in mammals, vertebrates and some invertebrates
(b) observed at a pressure above the atmospheric pressure
(c) It occurs under the influence of hydrostatic pressure
(d) It occurs under the influence of osmotic pressure
11. Process of guttation in plants for getting rid of excess water is a type of
(a) Imbibition (b) Filtration (c) Osmoregulation (d) Diffusion
12. Internalization of insulin in a cell occurs via the process of
(a) Reverse endosmosis (b) phagocytosis (c) pinocytosis
(d) receptor mediated endocytosis
13. The process of the transfer of a solute across a membrane from anode to cathode is known as _____
a) Metaphoresis b) Anaphoresis c) Cataphoresis d) Electrophoresis
14. Colloidal particles possess the property of attracting and holding other substance on their surface. The substance that sticks on the surface of colloid is called _____
a) Adsorbent b) Cation c) Anion d) Adsorbate
15. Freundlich's adsorption isotherm may be written as:
a) $a = kC^{1/n}$ b) $a = kC^{1/v}$ c) $a = k^{1/n}C$ d) $a = k^{1/v}C$
16. Tangential force per unit area per unit velocity gradient is called as _____
a) Hydrostatic pressure b) Surface tension c) Viscosity d) Coefficient of viscosity
17. "Substances which lower the surface tension become concentrated in the surface layer, whereas substances which increase surface tension, are more concentrated within the interior of the liquid than in the surface" This principle is given by:
a) Langmuir b) Freundlich c) Gibbs-Thomson d) Newton
18. Dynes-Second/ sq. cm is unit of _____
a) Viscosity b) Adsorption c) Surface tension d) Hydrotropy
19. The force of attraction between molecules of the different substance is called:
a) Adhesion b) Adsorption c) Absorption d) Cohesion
20. Reciprocal of the coefficient of viscosity ($1/\eta$) is called _____
a) Diffusion b) fluidity c) Surface tension d) Hydrotropy

Charotar University of Science and Technology

CHARUSAT Campus, Changa – 388 421.

M Sc (Nano Science and Technology)

SEMESTER II EXAMINATION 2011

NSIT 711 - Elective I: Biophysical Principles

Date: 6th May 2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Write answers for Section I and Section II of **Part B** in separate answer sheets.
3. Use of cell phones is strictly prohibited.
4. Use of non-programmable calculators may be allowed if needed.

Part B

Total Marks: 50

Section I

- Q1** A Give Difference between Absorption and Adsorption [4]
OR
A What is Reabsorption? Enlist the sites of reabsorption and discuss tubular reabsorption [4]
- B Give comparison of Physical adsorption and Chemisorption [3]
OR
B Write short note on Active absorption [3]
- C Briefly discuss Intestinal Absorption Vs. Absorption in the Alveoli [3]
- Q2** A Write short note on Ostwald Viscometer [4]
OR
A Discuss on molecular theory of Viscosity [4]
- B Explain following (Any Two)
a) Factors influencing the viscosity of Liquids [3]
b) Biological importance of Surface tension [3]
c) Capillary action [3]

Section II

- Q3** A Explain Fick's Law of diffusion. List the factors influencing it. [4]
B Differentiate between facilitated diffusion and diffusion [3]
OR
B Describe the factors affecting the rate of transport by facilitated diffusion [3]
- C Define facilitated diffusion. Explain using the example of sodium dependent glucose co-transport [3]
OR
C Explain the effect of concentration difference on diffusion of molecules across the membrane using neat labeled diagram [3]

P.T.O.

- [6]

OR

[4]

[4]

[6]

[4]

[4]